UNDER REVIEW

RESCIENCE C

Replication / Machine Learning

[Re] Accuracy is Not All You Need on Current Small Instruction-Tuned Models

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Abstract We reproduce the central finding of Dutta et al. (2024), that quantizing a language model can change a large fraction of its individual answers while leaving benchmark accuracy almost unchanged, and we test whether it holds on a newer and smaller generation of models. Using the original default bitsandbytes 8-bit and 4-bit configurations and metric definitions, we evaluate five current small instruction-tuned models (Qwen2.5-1.5B, Qwen2.5-3B, Llama-3.2-3B, Gemma-2-2B, and Phi-3.5-mini) on ARC-Challenge, MMLU, and five-option CommonsenseQA, and we add the widely used NF4 4-bit configuration as an extension. The core result reproduces. Flips are substantial, at 12 to 21 percent on 4-bit MMLU, and 4-bit quantization produces more flips than 8-bit in all 15 model-benchmark conditions. The direction of the flips also reproduces once the base rates of a multiple-choice task are taken into account. Incorrect answers change more often than correct ones in every condition, and changed incorrect answers land on the correct option far more often than chance, with a recovery rate on 4-bit MMLU of 47.5 percent against one third by chance and 43 to 60 percent in the original. The original margin explanation is supported, since correct answers with a small gap between the top two options change far more often than those with a large gap, although on these confident models many flips still come from large-margin answers. The claim that does not hold is accuracy preservation. Under the default 4-bit configuration, these models lose up to 7.8 points of accuracy, and on ARC-Challenge every model loses more accuracy than any model in the original study. Under NF4, the flips are about a third lower and the accuracy loss is about half as large. Repeated runs give identical probabilities, so the measured flips are not run-to-run noise.

A replication of Abhinav Dutta, Sanjeev Krishnan, Nipun Kwatra, and Ramachandran Ramjee. Accuracy is Not All You Need. Advances in Neural Information Processing Systems (NeurIPS), 2024.

1 Introduction

Dutta et al.[1] observed that quantizing a language model can leave its benchmark accuracy almost unchanged while altering a large fraction of its individual answers. They compared each quantized model against its full-precision baseline, counted the answers that changed between correct and incorrect, and called these changes flips. Because flips can reach double digits while accuracy moves by a point or less, accuracy alone does not reveal whether a quantized model behaves like the one it replaces. The finding is now widely cited and often treated as an established property of quantization.

The original study used 2023-era models, the Llama2-chat (7B, 13B, 70B) and Yi-chat (6B, 34B) families. These models are larger than, and were trained differently from, the small distilled models that practitioners quantize and run on consumer hardware today, which sit at or below three billion parameters. Whether the flips phenomenon behaves the same way on this newer generation had not been examined.

We replicate the bitsandbytes portion of Dutta et al.[1] on five current small instruction-tuned models across three multiple-choice benchmarks using the same default bitsand-

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Code is available at https://github.com/ab-ch90/quantization-flips-small-llms – DOI 10.5281/zenodo.21802395..
Data is available at https://github.com/ab-ch90/quantization-flips-small-llms.

bytes configurations as the original, and the core phenomenon holds. Quantization changes a substantial fraction of answers, and 4-bit quantization changes more answers than 8-bit quantization in every condition we test. The direction of the flips also replicates once the base rates of a multiple-choice task are taken into account. Initially incorrect answers change more often than initially correct ones, and a changed incorrect answer lands on the correct option far more often than chance, as the original study reports. With the full option probabilities saved, the margin explanation the original offers is also supported, as answers with a small gap between the top two options change far more often than answers with a large gap. The claim that does not hold on these models is accuracy preservation. Under the default 4-bit configuration, the models we test lose up to 7.8 points of accuracy, and on ARC-Challenge, which both studies evaluate zero-shot, every model loses more accuracy than any model in the original.

2 Scope and claims under test

We organize the replication around five claims from the original work:

- **C1.** Under quantization, benchmark accuracy is nearly preserved; the original reports changes of 0 to 2 percent on MMLU.
- **C2.** Flips are substantial even when accuracy is preserved.
- **C3.** Lower-bit quantization produces more flips than higher-bit quantization.
- **C4.** Correct-to-incorrect and incorrect-to-correct flips occur in roughly equal numbers, even though a task whose baseline answers are mostly correct would produce many more correct-to-incorrect flips by chance. This near balance is what allows accuracy to remain stable as flips accumulate.
- **C5.** Answers whose top two options are separated by a small probability gap (the top margin) change more often than answers with a large gap. Correct answers have larger margins and therefore change less often than incorrect ones.

We test the bitsandbytes 8-bit and 4-bit conditions on three of the original benchmark types, using the original metric definitions. We do not cover the GPTQ, AWQ, or SmoothQuant conditions or the generative-task analysis; the scope of this replication is the core quantization-flips result.

3 Methods

3.1 Models

We evaluate five current small instruction-tuned models, each of which fits in 16-bit precision on a single 8 GB consumer GPU, which makes the full-precision baseline computable locally: Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct [2], Llama-3.2-3B-Instruct [3], Gemma-2-2B-it [4], and Phi-3.5-mini-instruct [5]. These models span four families and two parameter scales below and around three billion. We chose breadth across families over breadth across sizes because the claim under test concerns quantization behavior in general.

3.2 Quantization

Each model is quantized with bitsandbytes [6] at 8-bit and 4-bit. Dutta et al.[1] state that their bitsandbytes conditions use the default parameters supported in Hugging Face Transformers, and we use the same defaults. The 8-bit condition is LLM.int8() with its default settings, and the 4-bit condition is the library's default 4-bit configuration, which uses the FP4 data type with 32-bit compute. These two conditions are the replication conditions, and all comparisons with the original use them. As an extension, we also evaluate the NF4 data type with 16-bit compute, a 4-bit configuration that is widely used in practice. We label the conditions 8-bit, FP4, and NF4, and we write 4-bit for the default FP4 condition. The 16-bit model serves as the reference for all comparisons.

3.3 Benchmarks

We evaluate on three multiple-choice benchmarks with objective answer keys: ARC-Challenge [7], MMLU [8], and CommonsenseQA [9]. These overlap with the original benchmarks and span science reasoning, broad knowledge, and commonsense reasoning. Each subset is fixed by dataset order, with no random sampling. For ARC-Challenge, we take the first 1,000 test items whose options are labelled A to D, which skips the items that use numeric labels or have three or five options. For MMLU, we take the first 125 test items from each of eight subjects chosen before running, two from each of its four categories (Appendix), which gives 975 items because one subject has only 100. For CommonsenseQA, we take the first 1,000 validation items with all five options, so that the task is the standard five-option benchmark.

3.4 Reading the answer

We read each model's answer from the logits of the option-letter tokens at the answer position, normalize them with a softmax over the four letters (five for CommonsenseQA), and take the option with the highest probability. Reading the letter probabilities directly avoids the measurement noise introduced by parsing generated text. For each item, we save the full probability distribution over the options, so the top margin of C5, the difference between the two largest option probabilities, is computed exactly. Each prompt is formatted with the model's own chat template. The Llama-3.2 template writes the current date into the system prompt unless a date is supplied, which would make its predictions depend on the day of the run, and we therefore fix that date to the template's own default of 26 July 2024.

3.5 Metric definitions

We follow the original metric definitions and extend them. Flips counts the answers that move from correct to incorrect together with those that move from incorrect to correct, and excludes answers that move from one wrong option to another. AllFlips adds those wrong-to-wrong transitions. We report both, together with the separate counts of correct-to-incorrect, incorrect-to-correct, and wrong-to-wrong transitions, and we test each accuracy change with an exact McNemar test.

To judge the direction of flips, we use two quantities that account for the base rates of a task with k options. The first is the recovery rate, the fraction of changed incorrect answers that land on the correct option. If a changed answer moved to one of the $k - 1$ other options at random, this fraction would be $1/(k - 1)$, which is one third for ARC-Challenge and MMLU and one quarter for CommonsenseQA. We test recovery

against this value with a one-sided binomial test and report Wilson 95 percent intervals. The second quantity is the chance ratio. If correct and incorrect answers changed at equal rates and every changed answer landed on a random other option, a baseline accuracy a would produce correct-to-incorrect and incorrect-to-correct flips in the ratio $(k-1)a/(1-a)$. At the accuracies in this study, that ratio ranges from 4.0 to 18.1, and a raw majority of correct-to-incorrect flips is therefore expected whenever accuracy exceeds $1/k$. We report the observed ratio for each condition with a bootstrap 95 percent interval over items.

3.6 Setting differences from the original study

Our setup differs from that of Dutta et al.[1] most of all in the models, which are a newer and more distilled generation. Two further differences bear on how directly the numbers compare. Our evaluation is zero-shot, whereas the original MMLU numbers are five-shot, and we evaluate a fixed subset of each benchmark instead of the full set. The original ARC-Challenge results are zero-shot, which makes that benchmark the closest point of comparison. For these reasons, we do not claim exact numerical replication of the reported flip rates. We claim instead that the phenomenon and its qualitative claims replicate, and we report where specific claims diverge.

3.7 Provenance and determinism

All runs used one NVIDIA RTX 3070 with 8 GB of memory, Python 3.12, PyTorch 2.6.0 with CUDA 12.4, Transformers 5.12.1, bitsandbytes 0.49.2, and Datasets 5.0.0. The commit hash of every model and dataset is pinned in the released code, and every result file records these revisions together with a hash of the item set, a hash of the rendered prompts, the item identifiers beside the predictions, the effective quantization configuration, and the software and hardware versions.

The flips metric compares a quantized run against a single full-precision baseline run and treats every difference as a consequence of quantization, which is valid only if both runs are stable. Recent work shows that language model inference can be nondeterministic even at temperature zero, mainly because of batch-dependent numerics in serving [10, 11]. We checked this directly, where for each of the five models at both 16-bit and FP4, we ran the first 300 ARC-Challenge items four times with nothing changed, and every run gave identical probabilities on every item, with a largest difference of exactly zero. Varying the batch size across four runs of Qwen2.5-3B at 16-bit changed a single item out of 300, a low-margin question that tipped under a small numeric change. Our main runs process one item at a time, so this batch effect does not enter them. As a further check across time, we compared the 16-bit, 8-bit, and NF4 predictions on ARC-Challenge and MMLU with those of an earlier run of the same pipeline about four months before. For Qwen2.5-1.5B, Qwen2.5-3B, Gemma-2, and Phi-3.5, every prediction matched. For Llama-3.2, about 3 percent of 16-bit predictions differed because the earlier run did not fix the date in the Llama prompt. All repeated runs are archived with the code.

4 Results

4.1 Substantial flips and dose-response (C2, C3)

The central finding that flips are substantial (C2) replicates. On MMLU, the default 4-bit condition produces Flips rates of 12.3 to 21.1 percent across the five models, with a mean of 15.7 percent (Figure 1). The original reports 4-bit MMLU Flips of 5.7 to 10.9 percent for its 2023 chat models (their Table 5), and every model we test lies above that range, consistent with the original's own finding that smaller models show more flips. Under NF4, the rates are lower, from 7.5 to 13.2 percent, and overlap the upper part of the original range. The size of the flips therefore depends on the 4-bit data type as well as on the model.

The dose-response claim (C3) holds in every condition. Both 4-bit configurations produce more flips than 8-bit quantization in all 15 model-benchmark conditions, and FP4 produces more flips than NF4 in all 15. Averaged over the three benchmarks, Flips rise from 3.3 to 5.1 percent at 8-bit to 6.5 to 10.0 percent under NF4 and 10.1 to 18.3 percent under FP4 (Figure 2). Full per-model numbers appear in Table 1.

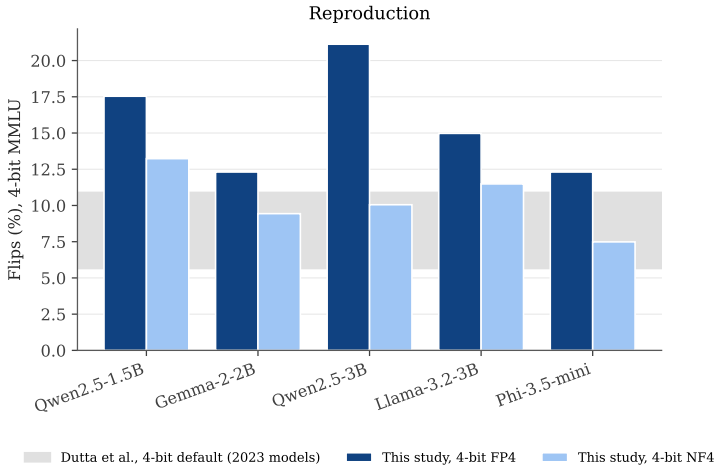


Figure 1. Replication of the flips finding on current small models. Bars show the 4-bit MMLU Flips rate for each model under the default FP4 configuration, which matches the original's condition, and under NF4. The shaded band marks the 5.7 to 10.9 percent range Dutta et al. report for their 2023 chat models under the default 4-bit configuration (their Table 5).

4.2 Direction of flips (C4)

In raw counts, correct-to-incorrect flips outnumber incorrect-to-correct flips. Pooled across the five models and three benchmarks under the default 4-bit condition, quantization produces 1,270 correct-to-incorrect flips against 680 in the other direction, a ratio of 1.87. This majority is expected and is not by itself evidence of a bias toward losing correct answers. Our baseline accuracies run from 57.0 to 85.8 percent, and at those accuracies the chance ratio defined above runs from 4.0 to 18.1. In every one of the 15 conditions, the observed ratio, between 0.97 and 3.05, is below its chance ratio, and the same holds for all 15 conditions at 8-bit and under NF4.

The two mechanisms that the original study identifies account for the gap. First, initially incorrect answers change much more often than initially correct ones. Pooled

Table 1. Full per-model, per-benchmark, per-precision results. Accuracy at full precision (fp16 Acc) and after quantization (Q Acc), the accuracy change (ΔAcc), Flips and AllFlips as percentages, and the directional breakdown of changed answers (C→I correct-to-incorrect, I→C incorrect-to-correct, W→W wrong-to-wrong) as counts. FP4 is the default 4-bit configuration used by the original, and NF4 is our extension. The benchmark subsets contain 1,000 ARC-Challenge, 975 MMLU, and 1,000 CommonsenseQA items.

Model	Benchmark	fp16 Acc	Q Acc	ΔAcc	Flips	AllFlips	C→I	I→C	W→W
8-bit									
Qwen2.5-1.5B	ARC-Challenge	75.3	76.2	+0.9	1.9	2.8	5	14	9
	MMLU	61.0	61.2	+0.2	4.3	5.8	20	22	15
	CommonsenseQA	73.4	73.5	+0.1	3.7	4.7	18	19	10
Gemma-2-2B	ARC-Challenge	74.4	74.5	+0.1	2.9	4.1	14	15	12
	MMLU	57.0	57.9	+0.9	3.6	5.2	13	22	16
	CommonsenseQA	69.8	69.6	-0.2	3.6	4.8	19	17	12
Qwen2.5-3B	ARC-Challenge	82.8	82.4	-0.4	3.8	5.1	21	17	13
	MMLU	65.4	64.0	-1.4	5.9	8.2	36	22	22
	CommonsenseQA	78.0	77.7	-0.3	5.7	6.9	30	27	12
Llama-3.2-3B	ARC-Challenge	71.1	71.1	+0.0	3.6	5.0	18	18	14
	MMLU	59.2	58.5	-0.7	5.4	8.6	30	23	31
	CommonsenseQA	69.8	70.1	+0.3	3.7	4.6	17	20	9
Phi-3.5-mini	ARC-Challenge	85.8	85.5	-0.3	2.9	3.4	16	13	5
	MMLU	70.8	69.7	-1.0	4.7	7.5	28	18	27
	CommonsenseQA	75.0	74.1	-0.9	5.9	7.6	34	25	17
4-bit FP4, default (replication condition)									
Qwen2.5-1.5B	ARC-Challenge	75.3	70.1	-5.2	12.8	17.6	90	38	48
	MMLU	61.0	54.6	-6.5	17.5	24.4	117	54	67
	CommonsenseQA	73.4	68.8	-4.6	13.2	17.1	89	43	39
Gemma-2-2B	ARC-Challenge	74.4	72.5	-1.9	9.1	11.7	55	36	26
	MMLU	57.0	57.2	+0.2	12.3	19.1	59	61	66
	CommonsenseQA	69.8	68.4	-1.4	9.8	13.2	56	42	34
Qwen2.5-3B	ARC-Challenge	82.8	75.0	-7.8	15.4	18.0	116	38	26
	MMLU	65.4	57.6	-7.8	21.1	28.9	141	65	76
	CommonsenseQA	78.0	70.2	-7.8	18.4	21.0	131	53	26
Llama-3.2-3B	ARC-Challenge	71.1	68.7	-2.4	11.0	13.5	67	43	25
	MMLU	59.2	56.1	-3.1	15.0	21.3	88	58	62
	CommonsenseQA	69.8	67.3	-2.5	11.1	13.8	68	43	27
Phi-3.5-mini	ARC-Challenge	85.8	83.0	-2.8	7.8	9.5	53	25	17
	MMLU	70.8	68.1	-2.7	12.3	16.8	73	47	44
	CommonsenseQA	75.0	71.7	-3.3	10.1	12.2	67	34	21
4-bit NF4, extension									
Qwen2.5-1.5B	ARC-Challenge	75.3	70.3	-5.0	8.6	11.1	68	18	25
	MMLU	61.0	56.6	-4.4	13.2	19.1	86	43	57
	CommonsenseQA	73.4	70.5	-2.9	8.3	12.3	56	27	40
Gemma-2-2B	ARC-Challenge	74.4	73.2	-1.2	6.4	8.7	38	26	23
	MMLU	57.0	57.0	+0.0	9.4	13.7	46	46	42
	CommonsenseQA	69.8	67.8	-2.0	6.2	9.6	41	21	34
Qwen2.5-3B	ARC-Challenge	82.8	80.2	-2.6	7.8	10.0	52	26	22
	MMLU	65.4	61.9	-3.5	10.1	14.9	66	32	47
	CommonsenseQA	78.0	74.5	-3.5	10.1	12.8	68	33	27
Llama-3.2-3B	ARC-Challenge	71.1	69.6	-1.5	8.7	11.9	51	36	32
	MMLU	59.2	57.7	-1.4	11.5	16.0	63	49	44
	CommonsenseQA	69.8	69.1	-0.7	8.3	10.5	45	38	22
Phi-3.5-mini	ARC-Challenge	85.8	84.2	-1.6	5.6	6.5	36	20	9
	MMLU	70.8	68.8	-1.9	7.5	10.7	46	27	31
	CommonsenseQA	75.0	73.7	-1.3	6.5	8.7	39	26	22

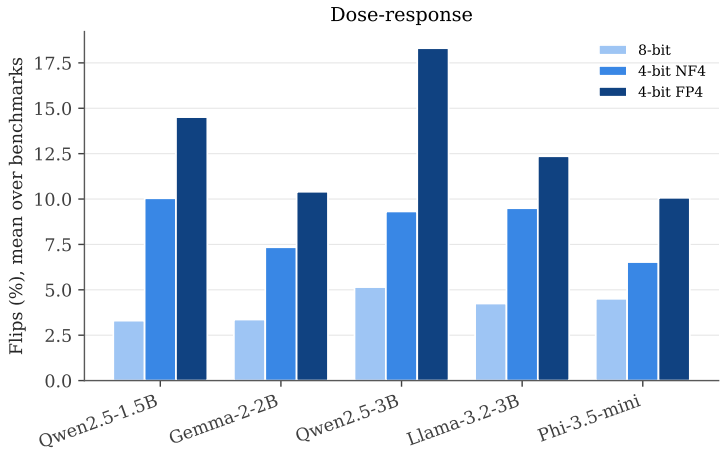


Figure 2. Mean Flips rate over the three benchmarks at 8-bit, NF4, and FP4. Both 4-bit configurations produce more flips than 8-bit for every model, and FP4 produces more than NF4.

under FP4, 30.1 percent of incorrect answers changed against 12.0 percent of correct answers, and incorrect answers changed more often in all 15 conditions, as the original reports in its Table 2. Second, a changed incorrect answer lands on the correct option far more often than chance. Pooled over models, the recovery rate is 55.9 percent on ARC-Challenge and 47.5 percent on MMLU, against one third, and 59.4 percent on five-option CommonsenseQA, against one quarter (one-sided binomial tests, $p < 10^{-12}$ for each benchmark). Recovery is above chance in all 15 conditions, and the difference is significant ($p < 0.05$) in every one of them (Figure 3). At 8-bit and under NF4, recovery is also above chance in all 15 conditions, and the difference is significant in 13 of the 15 for each.

Table 2 compares 4-bit MMLU directly with the original BnB W4A4 results, which we derive from their Tables 2 and 5. The recovery rates agree closely, as every model we test recovers between 44.6 and 51.6 percent of its changed incorrect answers within the original's range of 43 to 60 percent, and the pooled value is 47.5 percent. The ratio agrees less closely; when pooled over our models, it is 1.68 against 1.32 to 1.53 in the original, and the two Qwen models reach 2.17. These higher ratios come from a larger share of correct answers changing, 19.7 and 22.1 percent for the Qwen models against at most 12.8 percent in the original, which is the same accuracy loss reported under C1 below. Under NF4, the pooled ratio is 1.56, and the pooled recovery is 47.1 percent. Given the differences in shot count, benchmark coverage, and baseline accuracy, we read the comparison as agreement in direction and in recovery, with a somewhat higher ratio on our smaller models, and we conclude that C4 replicates.

4.3 Top margin and flip rate (C5)

Both parts of C5 replicate. First, correct answers have larger top margins than incorrect ones. On MMLU at full precision, the mean margin of correct answers is 0.62 to 0.97 across the five models, against 0.29 to 0.91 for incorrect answers, and it is larger for correct answers in every model, as in the original's Table 1. Second, answers with small margins change more often (Figure 4). Pooled over the three benchmarks under the default 4-bit condition, 43.2 to 64.0 percent of initially correct answers with a margin below 0.25 changed, depending on the model, against 1.9 to 16.1 percent of those with a margin of 0.80 or more. For every model, the correct answers that flipped had smaller

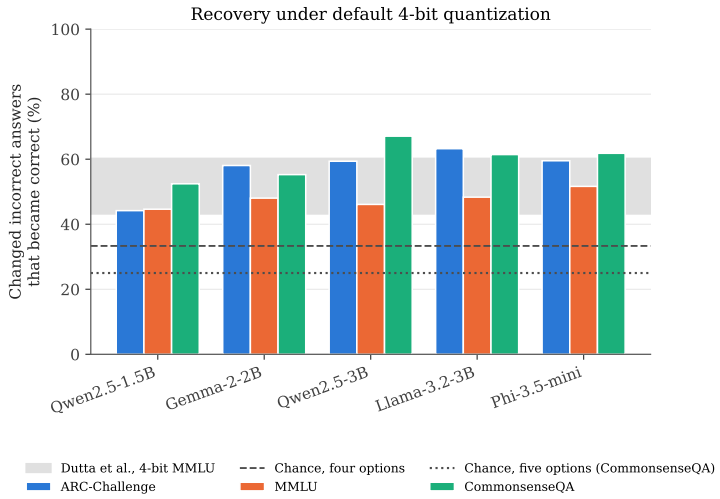


Figure 3. Recovery under the default 4-bit configuration, measured as the percentage of initially incorrect answers that changed and landed on the correct option, for each model and benchmark. The dashed line marks the one third expected on four-option ARC-Challenge and MMLU if a changed answer moved to a random other option, and the dotted line marks the one quarter expected on five-option CommonsenseQA. The shaded band marks the 43 to 60 percent range of the original BnB W4A4 MMLU results, derived from Dutta et al.'s Tables 2 and 5. All 15 conditions recover above chance.

Table 2. Direction of flips on 4-bit MMLU, compared with the original. Correct chg. and Incorrect chg. are the percentages of initially correct and initially incorrect answers that changed. Recovery is the percentage of changed incorrect answers that became correct, which is 33.3 percent under random landing. Chance ratio is $3a/(1-a)$, the C→I to I→C ratio expected if correct and incorrect answers changed at equal rates and landed on a random other option. Values for Dutta et al. are derived from their Tables 2 and 5 (arXiv version 1) and carry the rounding of those tables. Their setup is five-shot on the full benchmark, and ours is zero-shot on a 975-item subset. The last row pools our NF4 results for comparison.

Model	fp16 Acc	ΔAcc	Correct chg.	Incorrect chg.	C→I : I→C	Recovery	Chance ratio
Dutta et al., BnB W4A4, 5-shot, full MMLU							
Llama2-7B	47.2	-1.5	12.8	19.9	1.33	43.5	2.68
Llama2-13B	53.5	-1.3	8.7	16.9	1.39	43.2	3.46
Llama2-70B	63.2	-0.8	5.0	13.2	1.32	50.1	5.15
Yi-6B	63.0	-2.1	10.3	23.9	1.47	49.9	5.10
Yi-34B	74.9	-1.6	6.0	19.4	1.53	60.3	8.95
This study, bitsandbytes 4-bit default (FP4), zero-shot, 975-item subset							
Qwen2.5-1.5B	61.0	-6.5	19.7	31.8	2.17	44.6	4.70
Gemma-2-2B	57.0	+0.2	10.6	30.3	0.97	48.0	3.98
Qwen2.5-3B	65.4	-7.8	22.1	41.8	2.17	46.1	5.68
Llama-3.2-3B	59.2	-3.1	15.3	30.2	1.52	48.3	4.35
Phi-3.5-mini	70.8	-2.7	10.6	31.9	1.55	51.6	7.26
Pooled, FP4					1.68	47.5	
Pooled, NF4					1.56	47.1	

full-precision margins than those that did not (one-sided Mann-Whitney tests, $p < 10^{-60}$ for each model). The flip rate falls across the three margin groups for four of the five models. The exception is Qwen2.5-3B, whose two lower groups contain only 16 and 54 answers and flip at similar rates of 50.0 and 55.6 percent.

The margin explains which answers are most at risk but accounts for a smaller share of the flips than the original's account might suggest. Most correct answers from these models have a margin of 0.80 or more, from 60 percent of correct answers for Llama-3.2 to 97 percent for Qwen2.5-3B. A low flip rate applied to that large group still produces many flips, and the share of FP4 correct-to-incorrect flips that come from answers with a margin of 0.80 or more ranges from 10 percent for Llama-3.2 to 90 percent for Qwen2.5-3B. For Qwen2.5-3B, 16.1 percent of large-margin correct answers changed, which is also the model with the largest accuracy loss. On these models, the default 4-bit configuration perturbs the output distribution enough to overturn many answers the full-precision model held with near certainty.

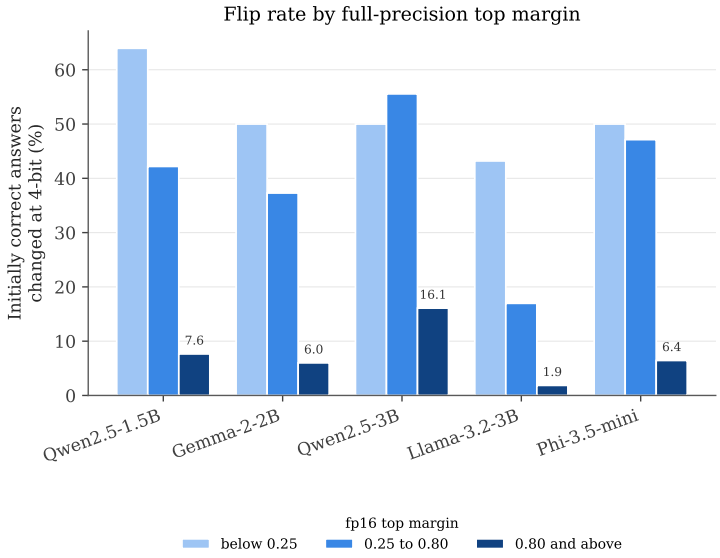


Figure 4. Percentage of initially correct answers that changed under the default 4-bit configuration, grouped by the full-precision top margin, the difference between the two largest option probabilities, and pooled over the three benchmarks. The flip rate is far lower for large-margin answers in every model.

4.4 Wrong-to-wrong transitions

The original study reports wrong-to-wrong transitions separately, in its AllFlips metric (their Figure 8 and Table 12), and excludes them from Flips to keep the metric applicable to generative tasks, where a wrong-to-wrong change is not well defined. On 4-bit MMLU, AllFlips exceeds Flips by a factor of 1.37 to 1.55 across our five models, within the factor of 1.26 to 1.56 implied by the original's Tables 5 and 12. A user comparing the full-precision and quantized models by AllFlips would see substantially more changed answers than Flips reports, as in the original.

4.5 Accuracy preservation (C1)

Accuracy preservation does not hold on these models under the default 4-bit configuration. The original reports MMLU accuracy changes of 0 to 2 percent, and its 4-bit BnB results range from -0.8 to -2.1 points. On 4-bit MMLU, Gemma-2 changes by $+0.2$ points, Phi-3.5 and Llama-3.2 lose 2.7 and 3.1 points, and Qwen2.5-1.5B and Qwen2.5-3B lose 6.5 and 7.8 points. The difference is clearer on ARC-Challenge, which both studies evaluate zero-shot. The original's 4-bit BnB accuracy changes on ARC-Challenge range from -0.9 to $+0.7$ points (their Table 9), while all five of our models lose more, from 1.9 points for Gemma-2 to 7.8 points for Qwen2.5-3B. The accuracy loss is significant (exact McNemar test, $p < 0.05$) in 12 of the 15 FP4 conditions, against none of the 15 at 8-bit, where accuracy changes by at most 1.4 points and C1 holds. Our models start from higher accuracy on ARC-Challenge (71 to 86 percent, against 44 to 55 percent in the original), which limits how directly the magnitudes compare, but the direction is consistent across all five.

The size of the loss depends on the 4-bit data type. Under NF4, the losses are roughly half as large, from 0.0 to 4.4 points on MMLU and from 1.2 to 5.0 points on ARC-Challenge, and Qwen2.5-3B loses 2.6 to 3.5 points instead of 7.8 on every benchmark. Even under NF4, every model loses more accuracy on ARC-Challenge than any model in the original. This result extends the original instead of contradicting it because the original's own results already show smaller models losing more accuracy and showing more flips than larger ones (their Table 5 and Section 4.1; for example, Llama2-7B loses 1.5 points on 4-bit MMLU against 0.8 for Llama2-70B), and the sub-3B models we test sit beyond that trend.

5 Discussion

On this generation of small instruction-tuned models, the flips phenomenon replicates, and so do the mechanisms the original study uses to explain it. A large share of answers change under quantization, heavier quantization changes more answers, incorrect answers change more often than correct ones, changed incorrect answers recover to the correct option well above chance, and small-margin answers are the most likely to change. A practitioner who quantizes one of these models and checks only accuracy will miss much of the behavioral change, as the original warned.

The direction of flips calls for care in interpretation. On any multiple-choice task with accuracy above chance, correct-to-incorrect flips will outnumber incorrect-to-correct flips under purely random perturbation, because more correct answers are available to break and a broken correct answer always becomes incorrect. A raw majority of correct-to-incorrect flips therefore says little on its own. The informative comparison is against the accuracy-dependent chance ratio, or equivalently between the recovery rate and its chance level. By those measures, the small models we test behave like the original's larger models.

The difference between the two generations lies in how much quantization costs. The smaller distilled models lose more accuracy, change more of their correct answers, and, under the default 4-bit configuration, change many answers they held with a large margin. A plausible account is that these models have less redundancy than the larger 2023 models, and the error that quantization introduces is therefore larger relative to the gaps between their option scores. We offer this as an interpretation and not a demonstrated mechanism because we measured which answers change and not why the weights behave as they do. Independent of this work, Kübler et al.[12] argue that the accuracy thresholds used to declare quantization lossless are too loose once estimation error is

taken into account, which aligns with the larger accuracy losses we observe. A separate line of work on quantization and confidence [13] studies how the whole confidence distribution shifts, whereas our emphasis is on which answers flip and in which direction.

The two 4-bit configurations also differ more than their shared bit width suggests. On the same models and items, FP4 produces about 1.5 times the flips of NF4 and nearly twice the accuracy loss. Comparisons between quantization studies therefore need the full configuration and not only the bit width, which is why we report both and record the effective configuration with every result. For someone choosing a quantization level for a small model on consumer hardware, accuracy on a benchmark remains an incomplete guide. Reporting Flips, AllFlips, and the directional breakdown alongside accuracy, with the direction judged against the chance baseline, gives a more complete picture of what quantization costs.

6 Limitations

Our setup differs from that of Dutta et al.[1] in ways that prevent exact numerical replication. We evaluate zero-shot while the original MMLU results are five-shot, and we use fixed subsets in place of the full benchmarks; a version that matched the shot count and used the full benchmarks would make the comparison tighter. The original values in Table 2 are derived from rounded percentages and are approximate. The low-margin groups in our C5 analysis are small for the models that are most confident, so the flip rates of those groups are imprecise. We test only bitsandbytes at two bit widths, and whether the same pattern appears under GPTQ, AWQ, or other schemes remains open. We test five models below and around three billion parameters; size and family are confounded in our set, and this is therefore not a controlled scaling study. Our determinism control establishes that our own measurements are not contaminated by run-to-run noise on our hardware, where the floor is zero, but it says nothing about hosted or batched serving, where the floor can be larger. Finally, we measure which answers change and characterize them by margin, but we do not offer a mechanistic account at the level of weights or activations.

7 Conclusion

The core of Dutta et al.[1] replicates on current small instruction-tuned models under the original's default bitsandbytes configurations. Flips are substantial (C2), and lower-bit quantization produces more flips in every condition (C3). Judged against the base rates of a multiple-choice task, the direction of flips also replicates (C4), with incorrect answers changing more often than correct ones and changed incorrect answers recovering to the correct option at rates inside the original's range. The margin explanation (C5) replicates, since small-margin answers change far more often than large-margin ones, although on these confident models, many flips still come from large-margin answers. Wrong-to-wrong transitions add to the behavioral change in the same proportion as in the original. The claim that does not hold is accuracy preservation (C1), where under the default 4-bit configuration, these smaller models lose up to 7.8 points of accuracy, and on ARC-Challenge every model loses more accuracy than any model in the original study.

Appendix: MMLU subjects

For MMLU, we use a fixed set of eight subjects, two from each of MMLU's four categories, to span the benchmark's range of content instead of clustering in one area. The subjects, by dataset identifier, are `college_computer_science` and `high_school_physics` (STEM); `philosophy` and `professional_law` (humanities); `high_school_psychology` and `high_school_macroeconomics` (social sciences); and `professional_medicine` and `marketing` (other). The set was fixed before any results were examined and was not revised in response to the outcomes.

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